



SELECTION OF MOTHER PALMS AND SEED NUTS

At present, the capacity of coconut seed gardens is not sufficient to cater the needs of high potential seed materials required for the coconut sector in Sri Lanka. As an alternative measure to establish coconut plantations with high genetic potential, selection of high yielding mother palms and seed nuts has become an essential practice. The scientific base for selecting palms with high genetic potential is on the nut yield and copra out turn of nuts. It has been observed that in a population established with unselected material, 80 percent of the total yield is derived from a mere 20 percent of the palms. Therefore, selection of parent palms and nuts for seeds is obviously a very useful practice in establishing high yielding and uniformly bearing coconut plantations. Mother palm and plus palm are two concepts developed in identifying palms as seed parents.

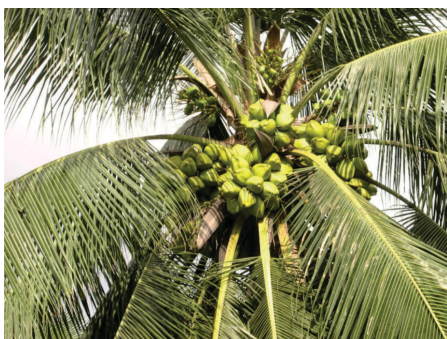


Figure 1: Well grown mother palm

A. Mother palm selection

Mother palms are selected according to the following procedure.

- a. Identify high yielding estates and select high yielding blocks within the estate.
- b. Select individual palms from the identified high yielding blocks based on desirable agronomic features as mentioned below (figure1).
 - i. Healthy, sturdy and straight stem with closely located leaf scars.
 - ii. Well grown leaves in a spherical orientation on the crown.
 - iii. The crown should be comprises of 25-30 fully opened fronds with bunches from button nuts up to mature nuts.
 - iv. Short and stout petioles capable of providing a solid support to bunchs.
 - v. Short and strong bunch stalks.
 - vi. Bunches of medium sized nuts in large numbers.

- c. In these palms, maintain separate records of nut yield in each pick and average weight of three husked nuts in 3-5 years.
- d. Select the best performing palms out of the initially selected pool according to the following standards.
 - i. Palms which yield more than 75 nuts / year.
 - ii. The minimum average weight of a husked nut should be more than 700 g (average of 3 nuts).
- e. Restrict the selection of palms to 5% to 10% in each block.

B. Plus palm selection

The mother palm selection is the best method for selecting parent palms. However, seeds supplied from seed gardens and mother palm section program are not sufficient to meet the increasing demand of seed nuts for the National Replanting Program (NRP). Therefore, an alternative method, plus palm selection was introduced to supplement the requirement of seed nuts for the NRP.

In Selecting plus palms, the following three main steps should be followed.

- a. Selection of high yielding estates and blocks within the estate.
- b. Selection of plus palms within blocks.
- c. Selection of seed nuts from plus palms.

1. Selection of high yielding blocks

- i. Select large blocks preferably not less than 5 acres.
- ii. Select blocks yielding over 3500 nuts/ac/yr or 60 nuts /palm/yr in recent five year period.
- iii. Select blocks with at least 55 bearing palms per acre (figure 2).
- iv. Select blocks with bearing palms between 15 to 45 years of age.



Figure 2: Selected high yielding block

2. Selection of plus palms within blocks

- i. Select hundred palms in the block randomly, harvest them and calculate the average number of nuts per palm.
- ii. Exclude all palms with undesirable characters (weak palms, tapered palms, Thambili and hybrids).
- iii. Select well grown palms with high nut yield and desirable agronomic characters.
- iv. Collect three mature nuts at random from each palm which has been selected on the basis of nut number and record their husked nut weight.
- v. Identify palms with at least 2100g of husked nut weight for the total of 3 nuts and mark them as plus palms.

3. Selection of seed nuts

It is preferred that harvesting of nuts of the marked plus palms are done separately to avoid mixing of nuts from unselected palms in loading and transportation. Only well matured nuts are selected as seed nuts. The best practice is to lay the nuts harvested from selected palms in nurseries immediately. If laying in nurseries is delayed nuts should be heaped and well covered with dry coconut fronds in order to minimize the loss of nut water.



Figure 3: Selected seed nuts

All immature, spoiled, empty, small and deformed nuts should be rejected before they are transported to nurseries. Extra- large nuts which are difficult to be laid in nurseries should also be rejected.

The selection of plus palms in the field is carried out jointly by Coconut Research Institute in the Coconut Cultivation Board. When the Coconut Research Institute is informed by the growers about the estate which could be selected on the above criteria, Coconut Research Institute could assist them to select plus palms in their estates.

Genetics and Plant Breeding Division
January 2018

